



EXPLAINABLE MODEL FOR DETECTION OF FETAL BRAIN ANOMALIES

Department of Computer Science and Engineering
PES University, RR Campus, Bengaluru - 560085.

PROBLEM STATEMENT

This project addresses critical challenges in medical image analysis, focusing on fetal ultrasound and brain MRI diagnostics. It aims to enhance fetal plane classification using Vision Transformers (ViTs) and explainability with Layer-wise Relevance Propagation (LRP), automate head circumference measurement while comparing models for abnormality detection, and improve brain tumor classification with ResNet models integrated with Grad-CAM for explainability.

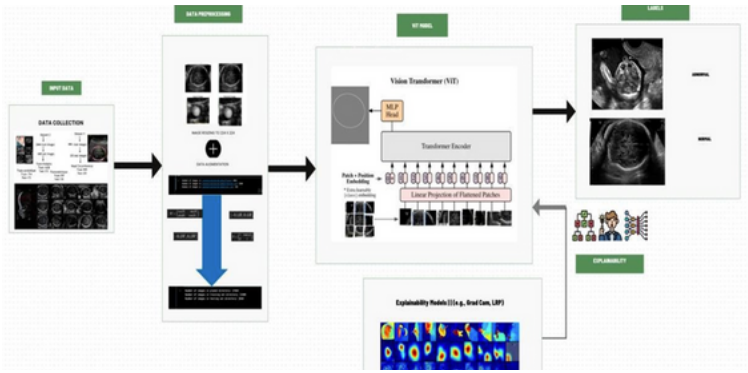
METHODOLOGY

- 1. Fetal Plane Classification:** Vision Transformers (ViT) integrated with Layer-wise Relevance Propagation (LRP) are used to classify fetal planes, ensuring interpretability of the model's decisions.
- 2. Head Circumference Calculation:** A UNet model segments relevant regions in ultrasound images, with LRP enhancing the understanding of segmentation decisions.
- 3. Anomaly Detection:** ResNet50, combined with Grad-CAM, is employed to detect fetal brain anomalies, providing heatmaps for explainability.

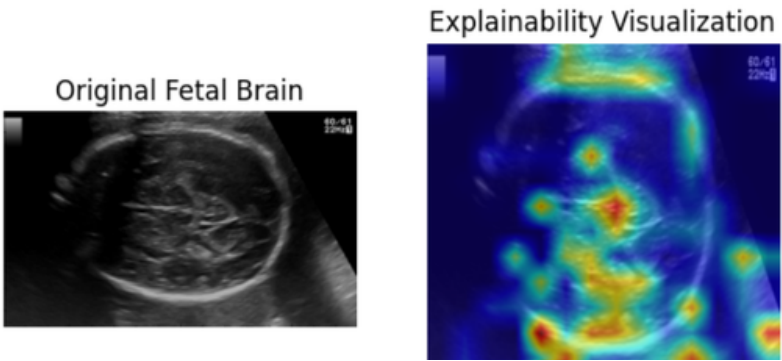
FEATURES

- The datasets utilized include ultrasound fetal images and MRI images for the third.
- For fetal plane classification, the ultrasound images were preprocessed to suit Vision Transformer (ViT).
- For brain tumor classification, MRI images were prepared for ResNet-based deep learning, facilitating robust model training and accurate predictions.
- The project requires a high-performance GPU (e.g., NVIDIA Tesla T4), deep learning libraries (PyTorch, OpenCV, torchvision), and explainability tools like Grad-CAM and LRP modules.
- The project offers automated detection, measurement, and classification tools for medical imaging, enhanced by model explainability for clinical reliability.

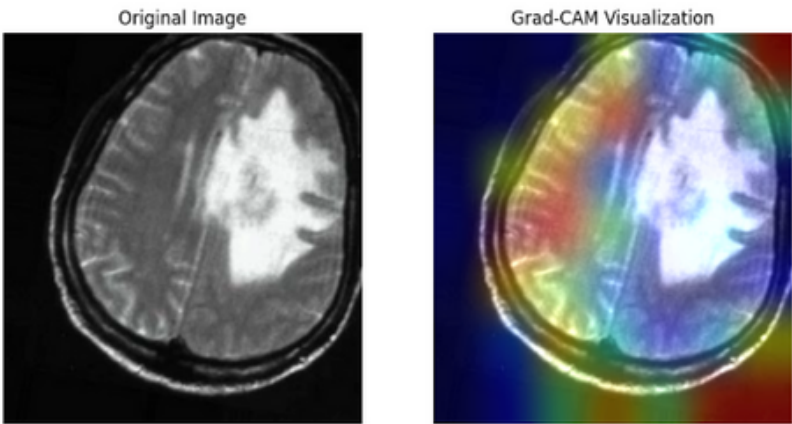
DESIGN APPROACH / METHODS



RESULTS AND DISCUSSION



Class	Precision (%)	Recall (%)	F1-Score (%)
Brain	0.97	0.99	0.98
Thorax	0.93	0.94	0.94
Abdomen	0.92	0.91	0.92
Cervix	0.99	0.99	0.99
Femur	0.92	0.91	0.94
Others	0.92	0.92	0.92



SUMMARY OF PROJECT OUTCOME

The project successfully met its initial objectives by developing three distinct models for medical image analysis. The use of Vision Transformers (ViT) and ResNet with explainability techniques (LRP and Grad-CAM) provided automated and interpretable solutions. The outcomes aligned with the initial estimates, demonstrating robust performance in classification accuracy and clinical interpretability, confirming that the project's goals were achieved within the expected scope.

CONCLUSION AND FUTURE WORK

The project successfully advanced automated medical image analysis by integrating state-of-the-art deep learning models with explainability techniques. The developed solutions offer promising potential for clinical applications, providing both high accuracy and interpretable results. Future work will focus on refining the models for greater precision, expanding to more diverse datasets, and exploring real-time applications in clinical environments to enhance decision-making and diagnostic accuracy.

REFERENCES

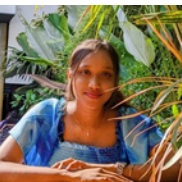
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Dr. Jayashree R
Mentor



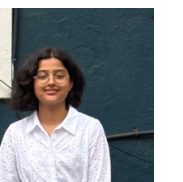
Sinchana S
PES1UG21CS596



Nisarga Kunder
PES1UG21CS392



Siri Gowri H
PES1UG21CS599



Shravani Hiremath
PES1UG21CS563